

Program : Diploma in Electrical and Electronics Engineering / Renewable Energy	
Course Code : 6299C	Course Title: Industrial Automation Lab
Semester : 6	Credits: 1.5
Course Category: Program Core	
Periods per week: 3 (L:0 T:0 P:3)	Periods per semester: 45

Course Objectives:

- To understand microcontroller and its programming methods.
- To summarize the interfacing of peripherals with microcontroller.
- To summarize the applications of PLC modules in electrical engineering.

Course Prerequisites:

Topic	Course code	Course name	Semester
Knowledge of motor drives		Dc machines	3
		Induction machines	4
Starters for induction motor		Industrial Electrical Engineering Lab	4
Knowledge of electronics		Analog & Digital Circuits	3
Basics of computer knowledge		Introduction to IT System Lab	1

Course Outcomes:

On completion of the course, the students will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Develop simple assembly language programs for 8051 microcontroller	14	Applying
CO2	Develop assembly language programs to interface the microcontroller with the peripherals.	6	Applying

CO3	Develop simple ladder programs for a standard PLC	9	Applying
CO4	Develop ladder programs for various applications and interface I/O devices with the PLC modules.	12	Applying
	Lab Exam	4	

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3			3			
CO2				3			
CO3	3			3			
CO4				3			

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcome	Description	Duration (Hours)	Cognitive level
CO1	Develop simple assembly language programs for 8051 microcontroller		
M1.01	Identify the Pin details of 8051 microcontroller	2	Understanding
M1.02	Familiarize with Instruction set of 8051	7	Understanding
M1.03	Summarize different menus available in simulator IDE software and practice their use.	1	Understanding
M1.04	Develop simple assembly language programs execute and verify the result for the following; 1. Various arithmetic operations. 2. Various data transfer operations. 3. Finding the largest value 4. BCD to Hex conversion	4	Applying

CO2	Develop Assembly language programs to interface the microcontroller with the peripherals.		
M2.01	Develop an assembly language programme to interface LEDs through port 1 including time delay.	2	Applying
M2.01	Develop an assembly language programme to control a stepper motor.	2	Applying
M2.01	Develop an assembly language programme to control a DC motor.	2	Applying
	Lab Exam 1	2	
CO3	Develop simple ladder programs for a standard PLC		
M3.01	Explain the block diagram of a PLC based system and distinguish its major components and functions.	1	Understanding
M3.02	Familiarize ladder programming and instruction set of a standard PLC	5	Understanding
M3.03	Familiarize a standard PLC kit and software	1	Understanding
M3.04	Develop simple ladder programs and verify the result 1. Various ON/OFF control with and without delay 2. Various logic gates	2	Applying
CO4	Develop ladder programs for various applications and interface I/O devices with the PLC modules.		
M4.01	Develop a ladder program and verify the result of a stair case light control	2	Applying
M4.02	Develop a ladder program and verify the result of a godown light control	2	Applying
M4.03	Develop a ladder program and verify the result of a traffic light control	2	Applying
M4.04	Develop a ladder program and verify the result of a direct on Line (DOL) starter	3	Applying
M4.05	Develop a ladder program and verify the result of a Star-delta starter.	3	Applying
	Lab Exam 2	2	

Text / Reference:

T/R	Book Title/Author
R1	Dunning, G., Introduction to Programmable Logic Controllers, Thomson /Delmar learning, New Delhi, 2005,ISBN 13 : 97814018842602.
R2	Jadhav, V. R., Programmable Logic Controller, Khanna publishers, New Delhi, 2017, ISBN : 97881740922813.
R3	Petruszella, F.D., Programmable Logic Controllers, McGraw Hill India, New Delhi, 2010, ISBN: 97800710673864.
R4	Hackworth, John; Hackworth, Federic, Programmable Logic Controllers, PHI Learning, New Delhi, 2003, ISBN : 97801306071885.
R5	Stenerson Jon, Industrial automation and Process control, PHI Learning, New Delhi, 2003, ISBN : 97801306189006.
R6	Mitra, Madhuchandra; Sengupta, Samarjit, Programmable Logic Controllers and Industrial Automation - An introduction, Penram International Publication, 2015, ISBN: 97881879721747.
R7	Boyar, S. A., Supervisory Control and Data Acquisition, ISA Publication, USA, ISBN: 9781936007097

Online Resources:

Sl.No	Website Link
1	https://nptel.ac.in/courses/108105062/
2	https://nptel.ac.in/content/storage2/courses/112103174/pdf/mod3.pdf
3	https://nptel.ac.in/courses/108/106/108106022/
4	https://www.youtube.com/watch?